



Faculty of Computer Science and Engineering

Bachelor Science (Hons) in Software Engineering

4th Year 2nd Semester – Final Examination - July 2026

SEN4308 – Artificial Intelligence MODEL Paper

Date :

Time :

Duration:

INSTRUCTIONS

A) This paper consists of **three (03) sections** on **eight (08) pages**, including this page. **Answer all questions** in Section A, Section B and Section C. Section A consists of fifteen (15) Multiple Choice Questions and carries 15 marks. Section B consists of four (04) structured questions and carries 45 marks. Section C consists of four (04) essay-type questions and carries 40 marks.

B) This examination accounts for **60%** of the overall course module assessment. The total maximum mark attainable for this paper is **100 marks**. The marks assigned for each question and sub-section are indicated in square brackets.

C) This is a **closed book examination**.

D) Mobile phones, smart watches, or any other communication devices are **not permitted**.

E) Clearly state any assumptions you make. If there is any doubt regarding the interpretation of a question, make a reasonable assumption and clearly mention it in your answer script.

F) Write your index number on each page of your answer script. Do not write your name.

Section A

1. Which of the following best describes Artificial Intelligence?
 - a) The study of only computer hardware
 - b) The development of systems that can perform tasks requiring intelligence
 - c) The process of storing data in tables
 - d) The design of computer cables
2. The Turing Test is mainly used to evaluate whether:
 - a) A machine can behave intelligently like a human
 - b) A computer can store more files
 - c) A program can run without syntax errors
 - d) A database can process transactions
3. Which school of thought in AI focuses on “doing the right thing”?
 - a) Acting humanly
 - b) Thinking humanly
 - c) Thinking rationally
 - d) Acting rationally
4. In a search problem, the goal test is used to:
 - a) Find whether the current state satisfies the objective
 - b) Calculate the size of memory
 - c) Store all past programs
 - d) Remove all repeated nodes
5. Which search algorithm explores nodes level by level?
 - a) Depth-First Search
 - b) Breadth-First Search
 - c) Hill Climbing
 - d) Genetic Algorithm
6. Uniform Cost Search expands the node with:
 - a) The lowest path cost
 - b) The highest heuristic value
 - c) The deepest level
 - d) The largest branching factor

7. In A* search, the evaluation function is:

- a) $f(n) = g(n) + h(n)$
- b) $f(n) = g(n) - h(n)$
- c) $f(n) = g(n) \times h(n)$
- d) $f(n) = h(n) / g(n)$

8. In Minimax algorithm, the MAX player tries to:

- a) Minimize the opponent's score only
- b) Maximize its own utility value
- c) Randomly choose any move
- d) Ignore the opponent's move

9. Alpha-Beta pruning is used to:

- a) Increase the number of searched nodes
- b) Remove unnecessary branches in a game tree
- c) Convert fuzzy values into crisp values
- d) Train a neural network

10. In fuzzy logic, membership values are usually in the range:

- a) -1 to 1
- b) 0 to 1
- c) 1 to 100
- d) 10 to 100

11. A fuzzy set differs from a crisp set because:

- a) It allows partial membership
- b) It does not use values
- c) It only accepts true or false
- d) It cannot represent uncertainty

12. In a Hidden Markov Model, emission probability means:

- a) Probability of moving from one hidden state to another
- b) Probability of an observation given a hidden state
- c) Probability of selecting a chromosome
- d) Probability of pruning a branch

13. A Multi-Agent System consists of:

- a) A single database
- b) Multiple interacting intelligent agents
- c) One fixed program only
- d) A single hardware device

14. In a Genetic Algorithm, the fitness function is used to:

- a) Measure the quality of candidate solutions
- b) Store images
- c) Remove all poor data
- d) Convert speech to text

15. Which of the following is an ethical concern in AI?

- a) Bias and discrimination
- b) Keyboard layout
- c) Screen resolution
- d) File compression

Section B

Answer all the questions. [45 marks]

Q1. Fill in the blanks using the keywords provided below. [2 × 5 = 10 marks]

- a) A search strategy that explores nodes level by level is known as _____.
- b) A search strategy that explores the cheapest path first is known as _____.
- c) A game-playing algorithm that assumes the opponent will play intelligently is known as _____.
- d) In fuzzy logic, converting a fuzzy output into a crisp value is called _____.
- e) In Genetic Algorithms, the process of randomly changing part of a chromosome is called _____.

Keywords:

Breadth-First Search, Uniform Cost Search, Minimax, Defuzzification, Mutation

Q2. Write short answers for the following. [5 × 3 = 15 marks]

- i) State three components of a search problem.
- ii) State three common types of uninformed search algorithms.
- iii) State three components of a Hidden Markov Model.
- iv) State three examples of AI applications in real life.
- v) State three ethical concerns related to AI.

Q3) Consider the following fuzzy logic scenario. [10 marks]

A smart air-conditioning system controls fan speed based on room temperature and humidity.

The input variables are:

- Temperature
- Humidity

The output variable is:

- Fan Speed

Answer the following:

- i) Identify three suitable fuzzy sets for temperature. [2 marks]
- ii) Identify three suitable fuzzy sets for humidity. [2 marks]
- iii) Identify three suitable fuzzy sets for fan speed. [2 marks]

iv) Write two fuzzy IF-THEN rules for this system. [4 marks]

Q4) Consider the following Hidden Markov Model scenario. [10 marks]

A smart healthcare monitoring system observes a patient's heart rate, body temperature and oxygen level. The actual health condition of the patient is not directly visible.

Answer the following:

i) Identify possible hidden states. [2 marks]

ii) Identify possible observations. [2 marks]

iii) Explain the meaning of transition probability in this scenario. [3 marks]

iv) Explain the meaning of emission probability in this scenario. [3 marks]

Section C

Answer all questions. [4 × 10 = 40 marks]

Q1. Predicate logic is used in Artificial Intelligence to represent facts, rules and reasoning about real-world situations. Consider a predicate logic-based system used to evaluate whether a robot is autonomous.

Given the paragraph:

“If a robot is autonomous, then it can make independent decisions. RoboX does not make independent decisions. RoboX only follows pre-programmed instructions.”

Answer the following: [10 marks]

a) Write the paragraph using suitable Predicate Logic expressions. [3 marks]

b) Convert the Predicate Logic expressions into Conjunctive Normal Form. [3 marks]

c) Hence prove that RoboX is not autonomous. [4 marks]

Q2. A robot is placed in a simple environment. The robot starts from node S and must reach node G. The possible movements are given below:

$S \rightarrow A, B$

$A \rightarrow C, D$

$B \rightarrow E, F$

$C \rightarrow G$

$D \rightarrow G$

$E \rightarrow G$

$F \rightarrow G$

Answer the following: [10 marks]

- a) Draw the state-space tree starting from node S. [2 marks]
- b) By disregarding repeated nodes in the OPEN list, construct the OPEN and CLOSED lists for Breadth-First Search until the goal is found. Clearly state the traversal and final path. [4 marks]
- c) By disregarding repeated nodes in the OPEN list, construct the OPEN and CLOSED lists for Depth-First Search until the goal is found. Clearly state the traversal and final path. [4 marks]

Q3. A smart hospital system uses Artificial Intelligence to coordinate healthcare services. The system includes several intelligent agents such as patient monitoring agents, medical robots, doctor-support agents, nurse-alert agents, pharmacy agents, and emergency response agents.

The patient monitoring agents collect data such as heart rate, body temperature, oxygen level, and blood pressure. Medical robots deliver medicine and equipment. Doctor-support agents analyse patient data and provide recommendations. Emergency response agents generate alerts when a patient's condition becomes critical.

Answer the following questions. [10 marks]

- a) Define an intelligent agent and explain how a patient monitoring agent can perceive and act in the smart hospital environment. [2 marks]
- b) Identify four agents that can exist in the smart hospital system and briefly explain the role of each agent. [3 marks]
- c) Explain how communication and cooperation among agents can improve patient care in the smart hospital system. [2 marks]
- d) Discuss three challenges that may occur when implementing a Multi-Agent System in a smart hospital environment. [3 marks]

Q4. Suppose you are requested to design and develop an AI solution for a tea plantation. Drones are used to monitor plants and capture images of tea leaves. The images are analysed to detect diseases, determine fertilizer requirements and support tea plucking decisions.

Answer the following questions.

- a) Explain why tea plantation domain experts are required for this AI project. [3 marks]
- b) Identify suitable AI techniques that can be used in the proposed solution. [4 marks]
- c) Discuss ethical or practical issues that should be considered when deploying this AI solution.

[3 marks]

***** END OF THE EXAM PAPER *****